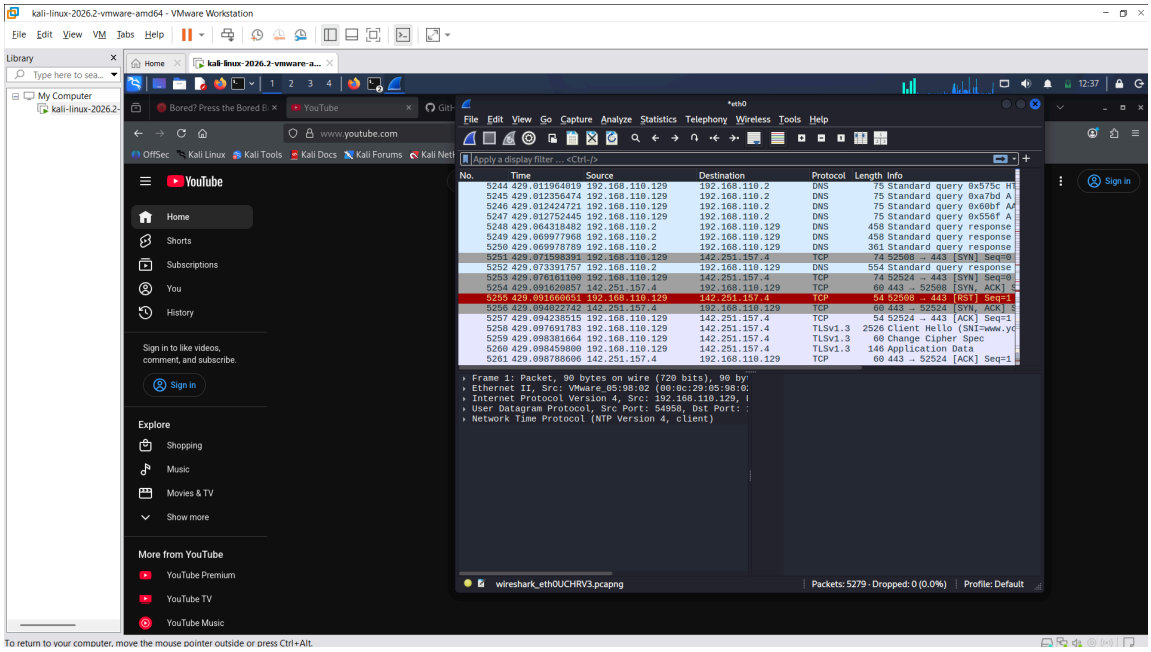
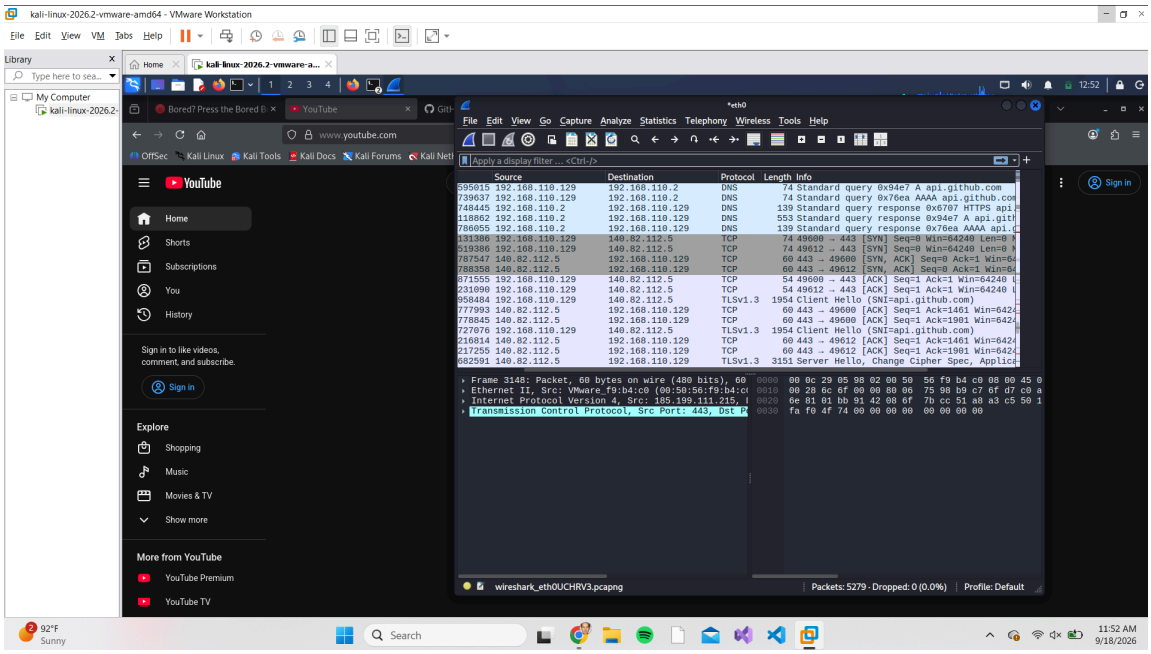
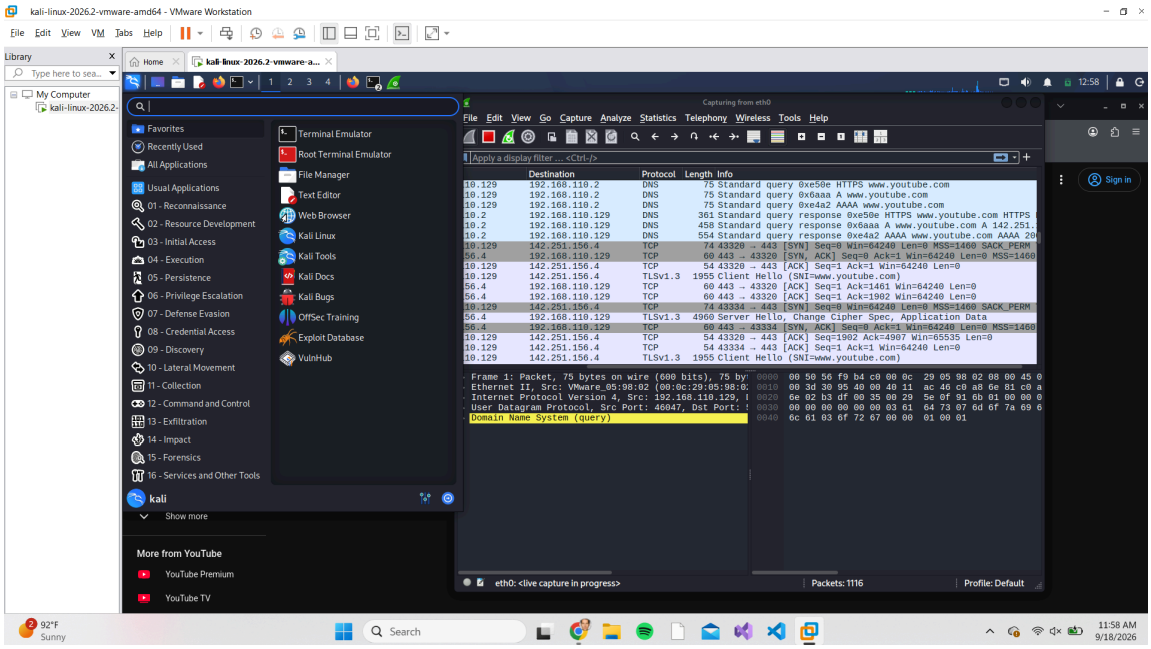
1. a) 1-(1-p)^h+d
   b) (CD/)H+D))(1-P)^H+D
2. a) 2Mb/s 2,000,000 b/s 8,000/2,000,000 = 4ms transmission delay + 2ms propagation delay 1Mb/s = 1,000,000 b/s 8000/1000000 = 8ms transmission delay + 1ms propagation delay total delay = 15ms total delay
   b) 2ms + 1ms + 2000/1000000 = 2ms + 1 = 6ms for 1st packet, and add slower bandwidth per other packet so + 2ms + 2ms + 2ms = 12ms total delay
3. a) Each object and image requires one HTTP request and response each, so the total number of requests is 4 and the total number of responses is also 4, when using non-persistent HTTP each object gets its own TCP connection, taking one RTT so the number of and each response and request pair takes one RTT, so it would be 8T, for persistent TCP we only need to establish one connection, therefore TCP only takes one RTT so it would be 5T, therefore 3T is saved using persistent HTTP in this scenario.
   b) Yes a response can have an empty body in the case of sending an error, such as error 404 because there is no body to return as the information wasn't found.
4. a) DNS, TCP, and TLSv1.3



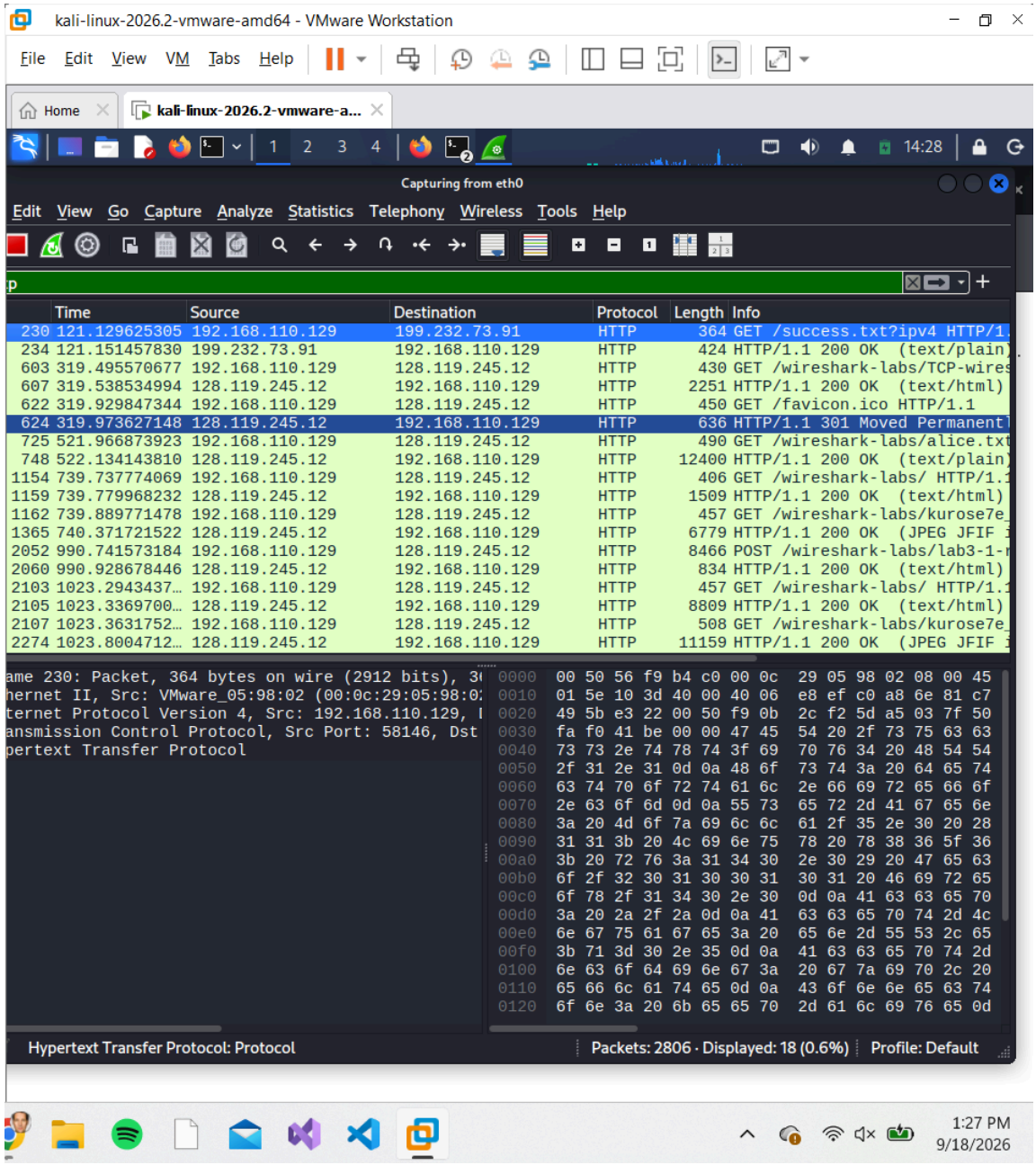
   b)





The IP of my host is 192.168.110.129
The IP of github is 140.82.112.5
The IP of youtube is 142.251.156.4

5. a)



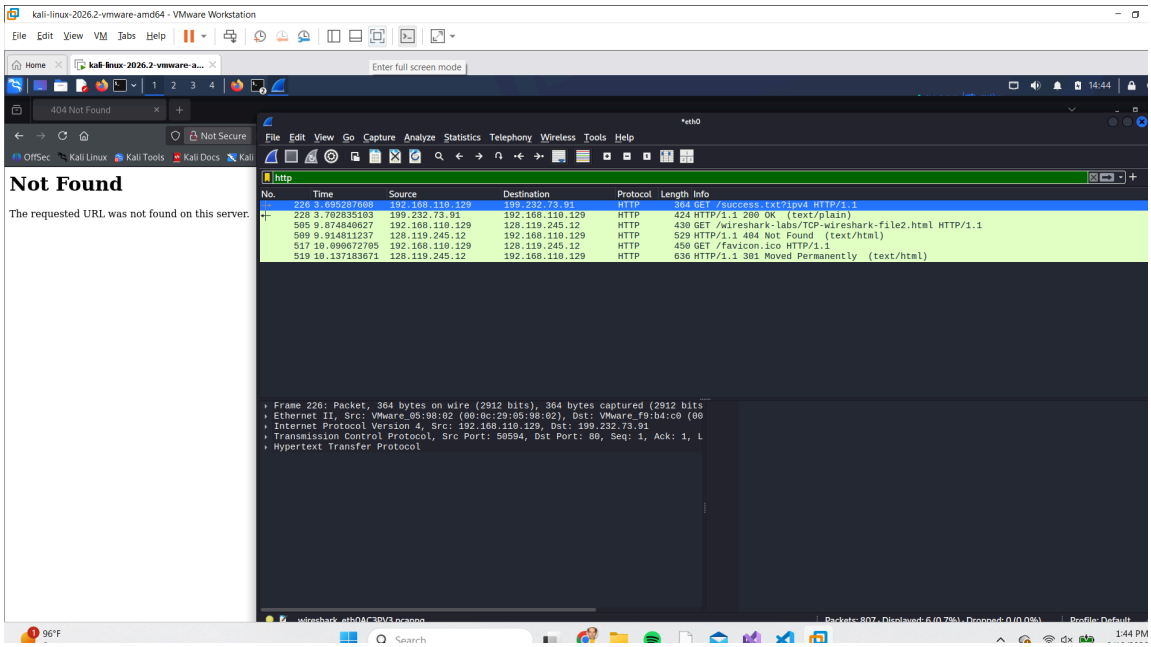
Time 121.1514 - 121.1296 = .0218

b) They are both running on HTTP1.1

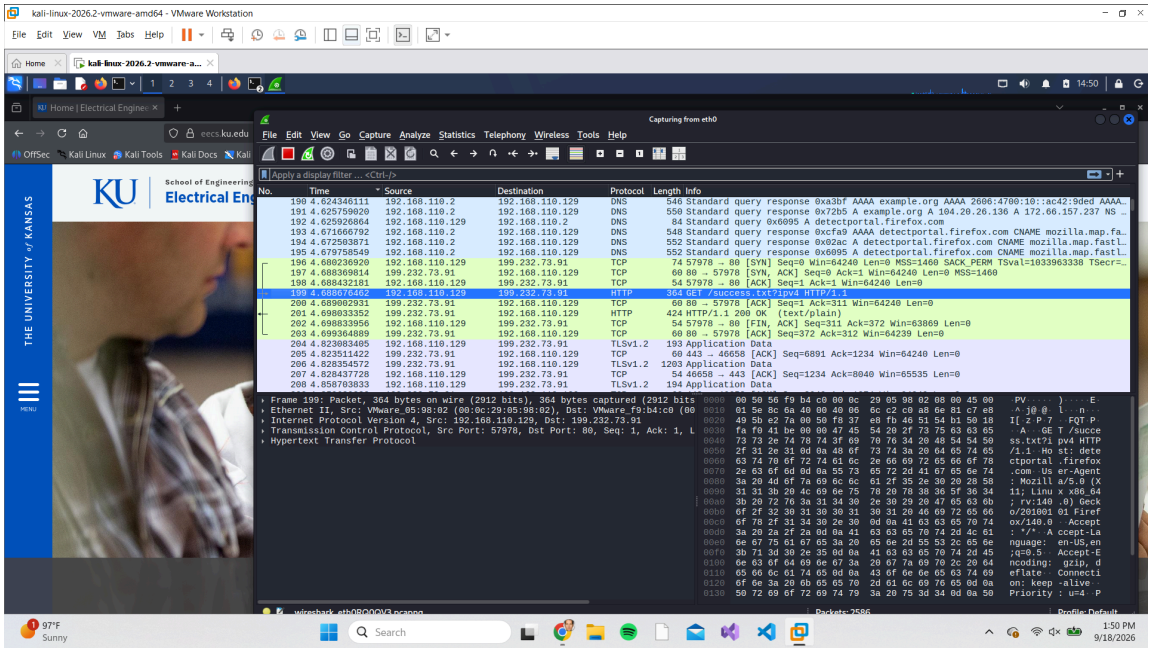
c) Host IP: 192.168.110.129 Web server: 128.119.245.12

d) 200

e) The error code is 404


kali-linux-2026.2-vmware-amd64 - VMware Workstation
404 Not Found
Not Found
The requested URL was not found on this server.
*eth0
http
No. Time Source Destination Protocol Length Info
226 3.695287608 192.168.110.129 199.232.73.91 HTTP 364 GET /success.txt?ipv4 HTTP/1.1
228 3.702835103 199.232.73.91 192.168.110.129 HTTP 424 HTTP/1.1 200 OK (text/plain)
505 9.874840627 192.168.110.129 128.119.245.12 HTTP 430 GET /wireshark-labs/TCP-wireshark-file2.html HTTP/1.1
509 9.914811237 128.119.245.12 192.168.110.129 HTTP 529 HTTP/1.1 404 Not Found (text/html)
517 10.090672705 192.168.110.129 128.119.245.12 HTTP 450 GET /favicon.ico HTTP/1.1
519 10.13718367 128.119.245.12 192.168.110.129 HTTP 636 HTTP/1.1 301 Moved Permanently (text/html)
Frame 226: Packet, 364 bytes on wire (2912 bits), 364 bytes captured (2912 bits
Ethernet II, Src: VMware_05:98:02 (00:0c:29:05:98:02), Dst: VMware_f9:b4:c0 (00
Internet Protocol Version 4, Src: 192.168.110.129, Dst: 199.232.73.91
Transmission Control Protocol, Src Port: 50594, Dst Port: 80, Seq: 1, Ack: 1, L
Hypertext Transfer Protocol

f)HTTP


kali-linux-2026.2-vmware-amd64 - VMware Workstation
Capturing from eth0
No. Time Source Destination Protocol Length Info
190 4.624346111 192.168.110.2 192.168.110.129 DNS 546 Standard query response 0xa3bf AAAA example.org AAAA 2606:4700:10::ac42:9ded AAAA
191 4.625759020 192.168.110.2 192.168.110.129 DNS 550 Standard query response 0x72b5 A example.org A 104.20.26.136 A 172.66.157.237 NS
192 4.625926864 192.168.110.129 192.168.110.2 DNS 84 Standard query 0x6095 A detectportal.firefox.com
193 4.671666792 192.168.110.2 192.168.110.129 DNS 548 Standard query response 0xcfa9 AAAA detectportal.firefox.com CNAME mozilla.map.fa
194 4.672503871 192.168.110.2 192.168.110.129 DNS 552 Standard query response 0x02ac A detectportal.firefox.com CNAME mozilla.map.fastl
195 4.679758549 192.168.110.2 192.168.110.129 DNS 552 Standard query response 0x6095 A detectportal.firefox.com CNAME mozilla.map.fastl
196 4.680236920 192.168.110.129 199.232.73.91 TCP 74 57978 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1033963338 TSecr=
197 4.688369814 199.232.73.91 192.168.110.129 TCP 60 80 → 57978 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
198 4.688432181 192.168.110.129 199.232.73.91 TCP 54 57978 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
199 4.688674442 192.168.110.129 199.232.73.91 HTTP 364 GET /success.txt?ipv4 HTTP/1.1
200 4.689002931 199.232.73.91 192.168.110.129 TCP 60 80 → 57978 [ACK] Seq=1 Ack=311 Win=64240 Len=0
201 4.698033352 199.232.73.91 192.168.110.129 HTTP 424 HTTP/1.1 200 OK (text/plain)
202 4.698833956 192.168.110.129 199.232.73.91 TCP 54 57978 → 80 [FIN, ACK] Seq=311 Ack=372 Win=63869 Len=0
203 4.699364889 199.232.73.91 192.168.110.129 TCP 60 80 → 57978 [ACK] Seq=372 Ack=312 Win=64239 Len=0
204 4.823083405 192.168.110.129 199.232.73.91 TLSv1.2 193 Application Data
205 4.823511422 199.232.73.91 192.168.110.129 TCP 60 443 → 46658 [ACK] Seq=6891 Ack=1234 Win=64240 Len=0
206 4.828354572 199.232.73.91 192.168.110.129 TLSv1.2 1203 Application Data
207 4.828437728 192.168.110.129 199.232.73.91 TCP 54 46658 → 443 [ACK] Seq=1234 Ack=8040 Win=65535 Len=0
208 4.858703833 199.232.73.91 192.168.110.129 TLSv1.2 194 Application Data
Frame 199: Packet, 364 bytes on wire (2912 bits), 364 bytes captured (2912 bits
Ethernet II, Src: VMware_05:98:02 (00:0c:29:05:98:02), Dst: VMware_f9:b4:c0 (00
Internet Protocol Version 4, Src: 192.168.110.129, Dst: 199.232.73.91
Transmission Control Protocol, Src Port: 57978, Dst Port: 80, Seq: 1, Ack: 1, L
Hypertext Transfer Protocol